

Project Overview

The project was aimed at developing a Python-based Habit Tracking Application that enables users to create, manage and monitor their habits. Object-Oriented Programming was used to structure the application into classes responsible for core habit operations, including adding, updating, deleting, and marking habits as completed. Habit data is stored persistently in JSON format, making the application modular, reusable, and easy to maintain. The application also applied Functional Programming principles to process and analyze habit data including calculating current and longest streak, generating habit summaries and completion analysis.

During the development of the project, various challenges and successful implementations were faced. These helped improve the overall outcome of project through continuous refinement.

a) Successful Implementations

- Separating the application into modules improved code organization, maintainability and testing. This helped in code organization and testing.
- Designing the CLI with tables and color-coded outputs improved readability and made it more user friendly.
- Testing the individual components helped identify issues early.
- Applying OOP principles helped with better organization and easier to extend.

b) Challenges Encountered

- Implementing streak calculations was challenging, especially ensuring the current streak restarted after a missed completion.
- Calculating the longest streak accurately from the completion history.
- Ensuring the habit data remained consistent when saving and loading from JSON.
- Handling cases where the user entered habit names or descriptions differed in capitalization from the stored JSON data.

c) How the challenges were resolved

- Current streak calculation was improved by checking the completion dates backwards and resetting the streak when a missed period is detected.
- Longest streak calculation was resolved by processing the completion history in chronological order to count consecutive completions and track the highest streak.
- JSON consistency was maintained by using the ISO date formatting and validating to ensure the saving and loading of habit data.

- For handling the capitalization issue, the text was converted to a consistent format (using the `.lower()`) before matching to with stored JSON data

My most valuable feature implemented is the accurate streak tracking system, which correctly calculates current and longest streaks accurately by analyzing the completion history. This gives users reliable feedback on their progress.

GitHub Repository

<https://github.com/EwubeE/Object-Oriented-and-Functional-Programming-with-Python.git>